Likert (lick-urt) or leave it: pairing landowner attitude data with carnivore resource selection data

Riecke, *University of Montana*

Blackfoot CHALLENGE

Better Rural Communities through Collaborative Conservation



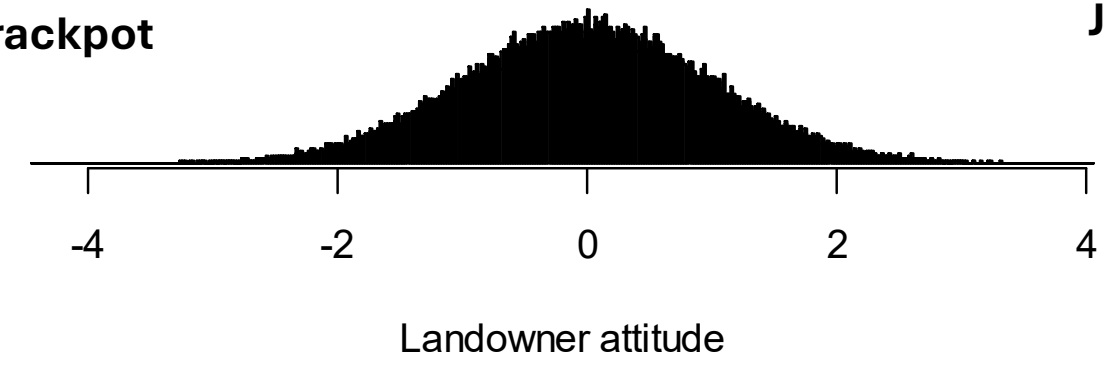
These are simulated [not real] data.



Ye ol' credulous crackpot

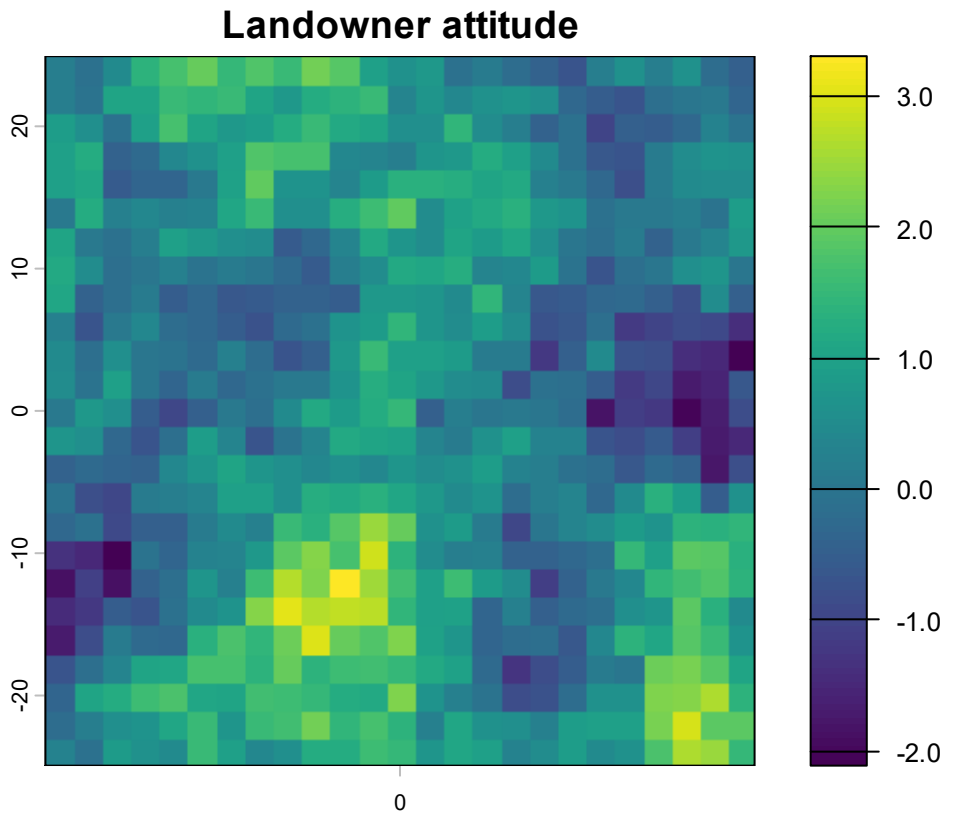


Jim Stone



But, we might imagine variation in landowner attitudes towards wildlife

We simulated variation in landowner attitudes

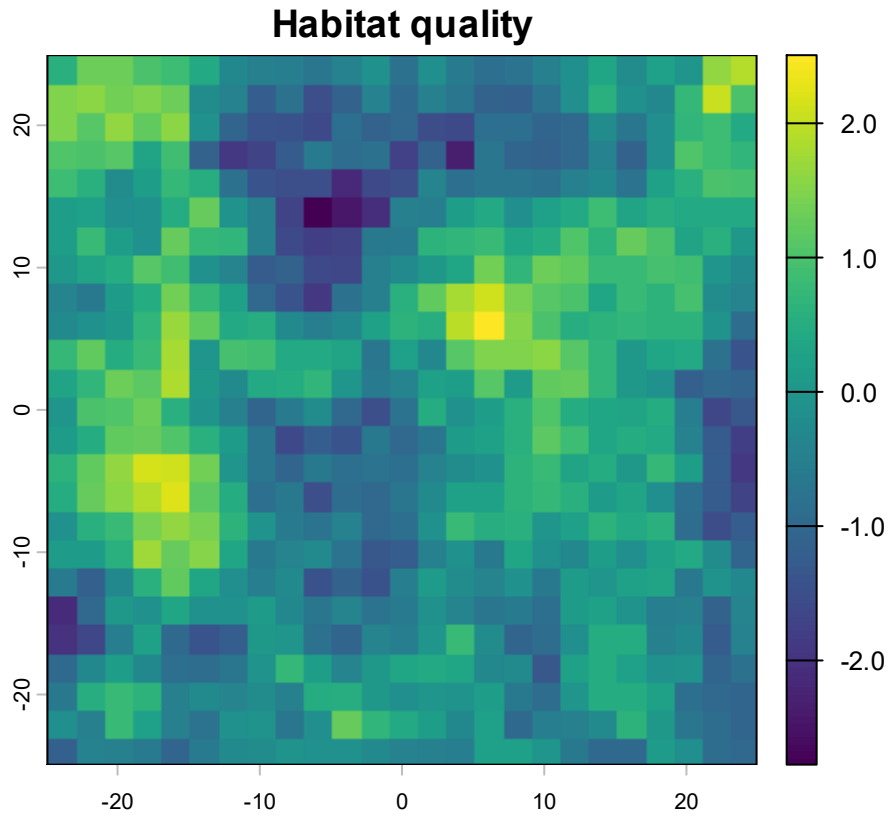






Gaussian random field

We also simulated variation in habitat quality

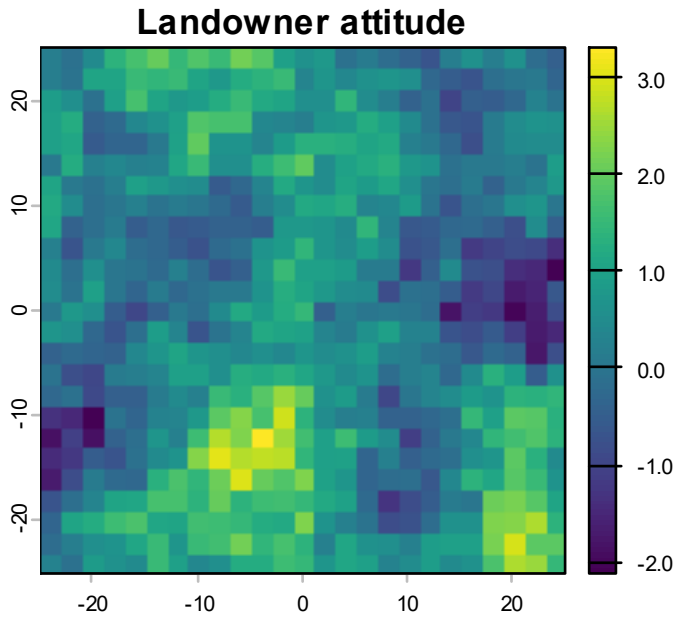


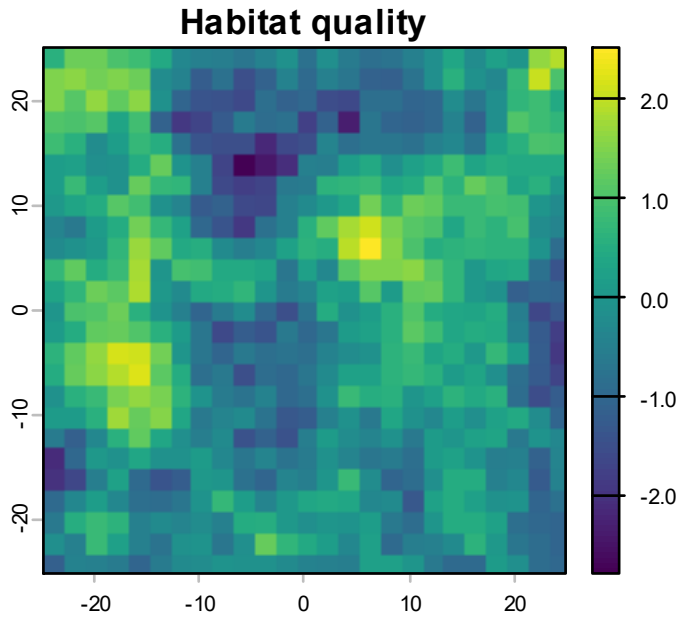


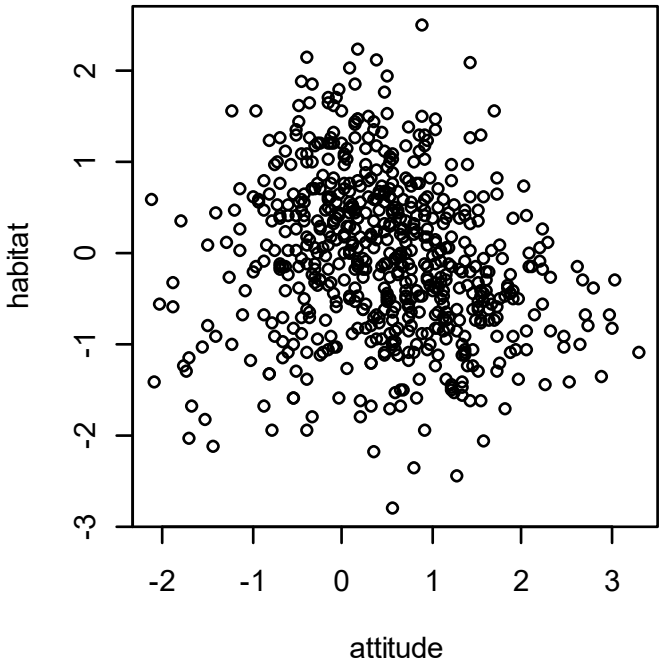


Gaussian random field

These variables were independent





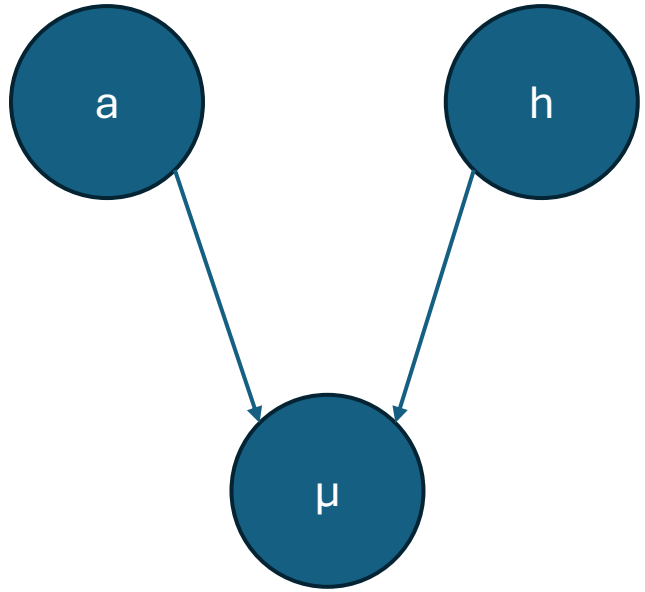


But they don’t have to be! See example 2!

We simulated resource selection as an inhomogeneous point process

$$\ln(\mu_i) = \alpha_1 + \alpha_2 \times a_i + \alpha_3 \times h_i$$

$$y_i \sim \text{Poisson}(\mu_i)$$



y: Number of unique bears using each cell?

Then we asked all 625 landowners a simple survey with three statements…

1. **I feel threatened by grizzly bears**
2. **Grizzly bears negatively affect my economic well-being**
3. **I think grizzly bears do not belong where people live**

Respondents were asked to respond on a scale from 1 to 5
(1. strongly disagree, 2. disagree, 3. neutral, 4. agree, 5. strongly agree)

Nesbitt et al. **(2023)** *Conservation Science and Practice*

So… these data are ordinal, or ordered categories

1. strongly disagree
2. disagree
3. neutral
4. agree
5. strongly agree

Ordinal data

- Scale of 1 to 10
- How ya feeling?
  - Grumpy, Fine, Ok, Good, Great, Best
- How’d it go?
  - Disaster, SNAFU, OK, Good, Great, Perfect

Other examples of ordinal data

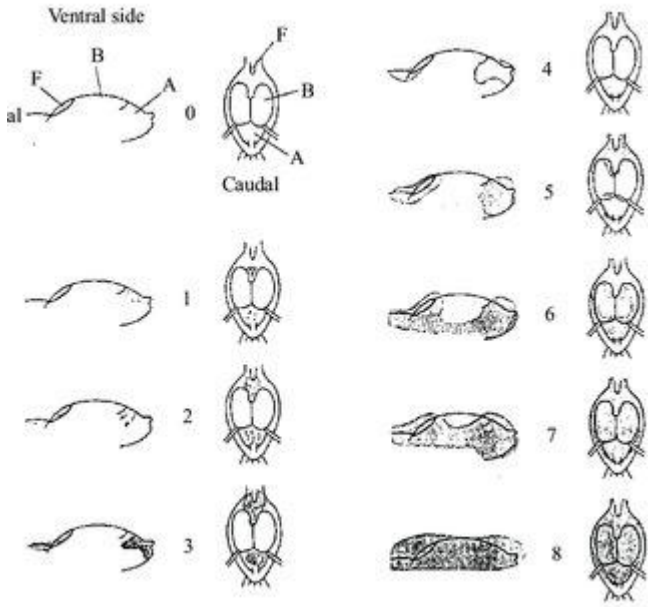
Credit Rating Scales by Agency, Long-Term

Moody's	S&P	Fitch	
Aaa	AAA	AAA	Prime
Aa1	AA+	AA+	High grade
Aa2	AA	AA	
Aa3	AA-	AA-	
A1	A+	A+	Upper medium grade
A2	A	A	
A3	A-	A-	
Baa1	BBB+	BBB+	Lower medium grade
Baa2	BBB	BBB	
Baa3	BBB-	BBB-	
Ba1	BB+	BB+	Non-investment grade speculative
Ba2	BB	BB	
Ba3	BB-	BB-	
B1	B+	B+	Highly speculative
B2	B	B	
B3	B-	B-	
Caa1	CCC+	CCC	Substantial risk
Caa2	CCC		Extremely speculative
Caa3	CCC-		Default imminent with little prospect for recovery
Ca	CC	CC	
	C	C	
C	D	D	In default
/			
/			

"Junk"

WOLFSTREET.com







We might expect answers to be correlated?

1. **I feel threatened by grizzly bears**
2. **Grizzly bears negatively affect my economic well-being**
3. **I think grizzly bears do not belong where people live**

If you feel threatened by grizzly bears, you might think they don't belong where people live

Nesbitt et al. **(2023)** *Conservation Science and Practice*

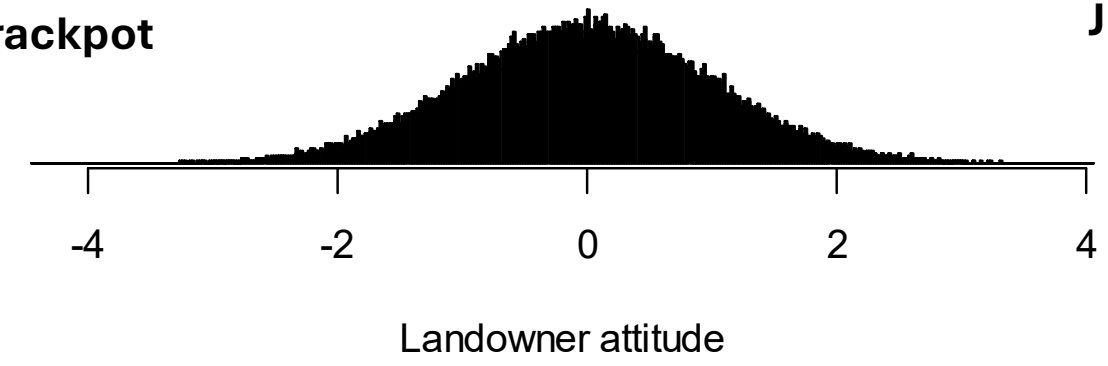
We simulated correlated responses to these questions



Ye ol' credulous crackpot



Jim Stone



Landowners with good attitudes disagreed (remember, simulated data)

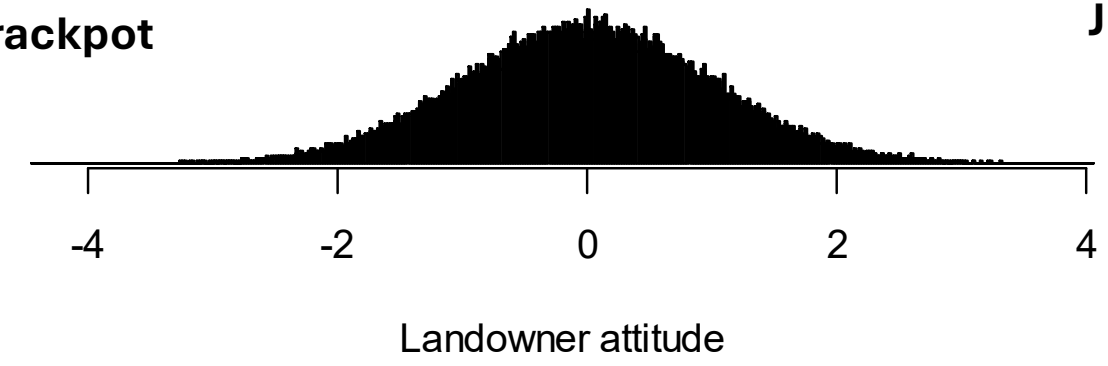
We simulated correlated responses to these questions



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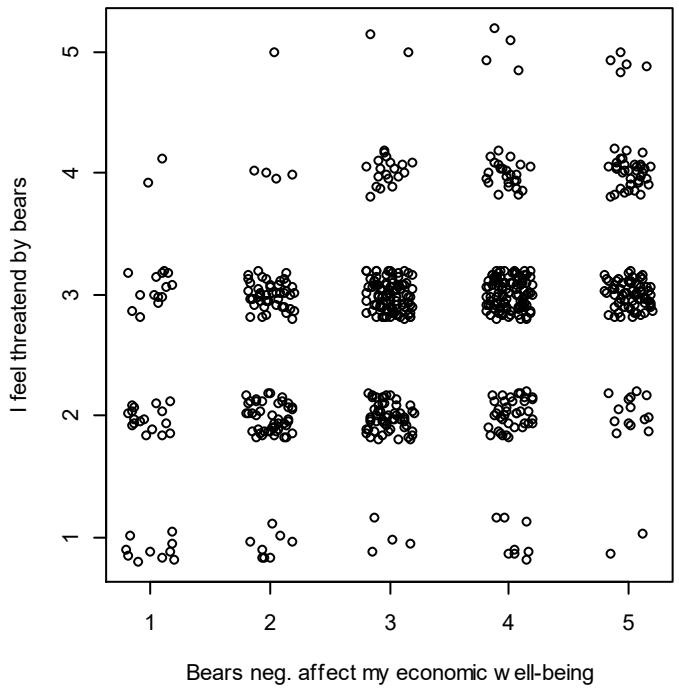


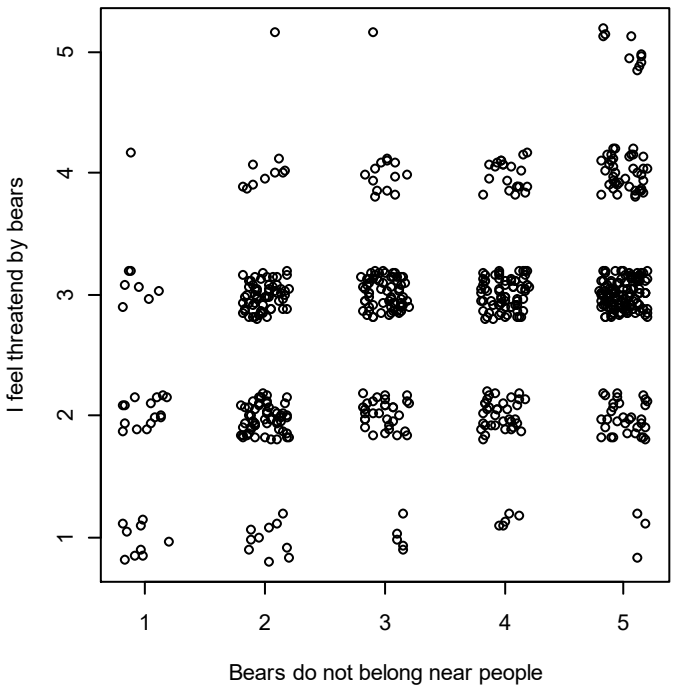
Jim Stone

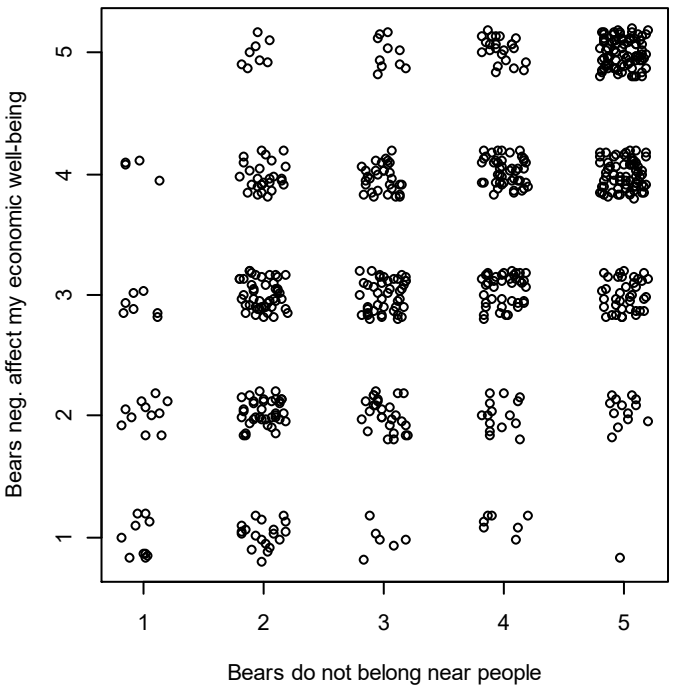


And vice versa (remember, simulated data)

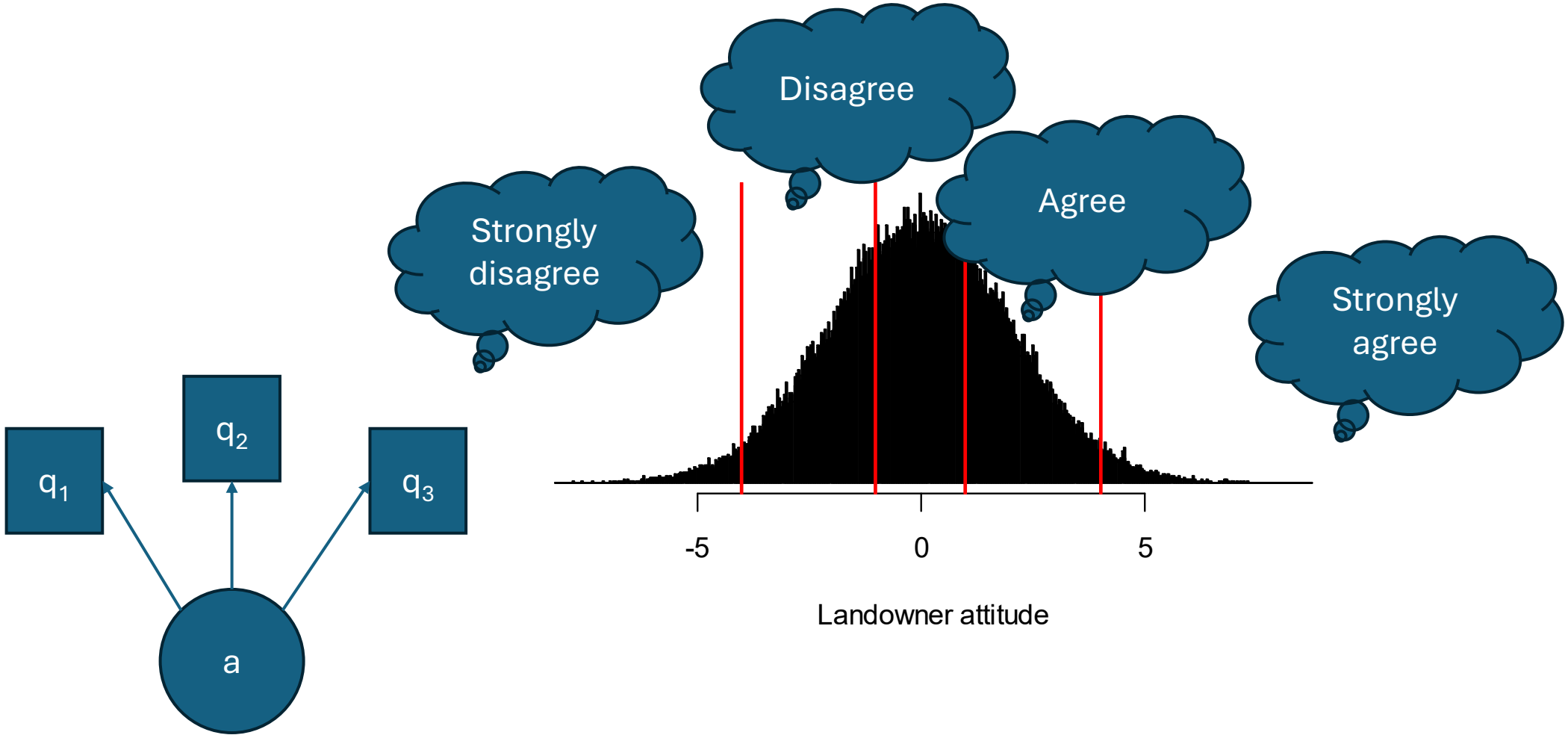
We simulated correlated responses to these questions





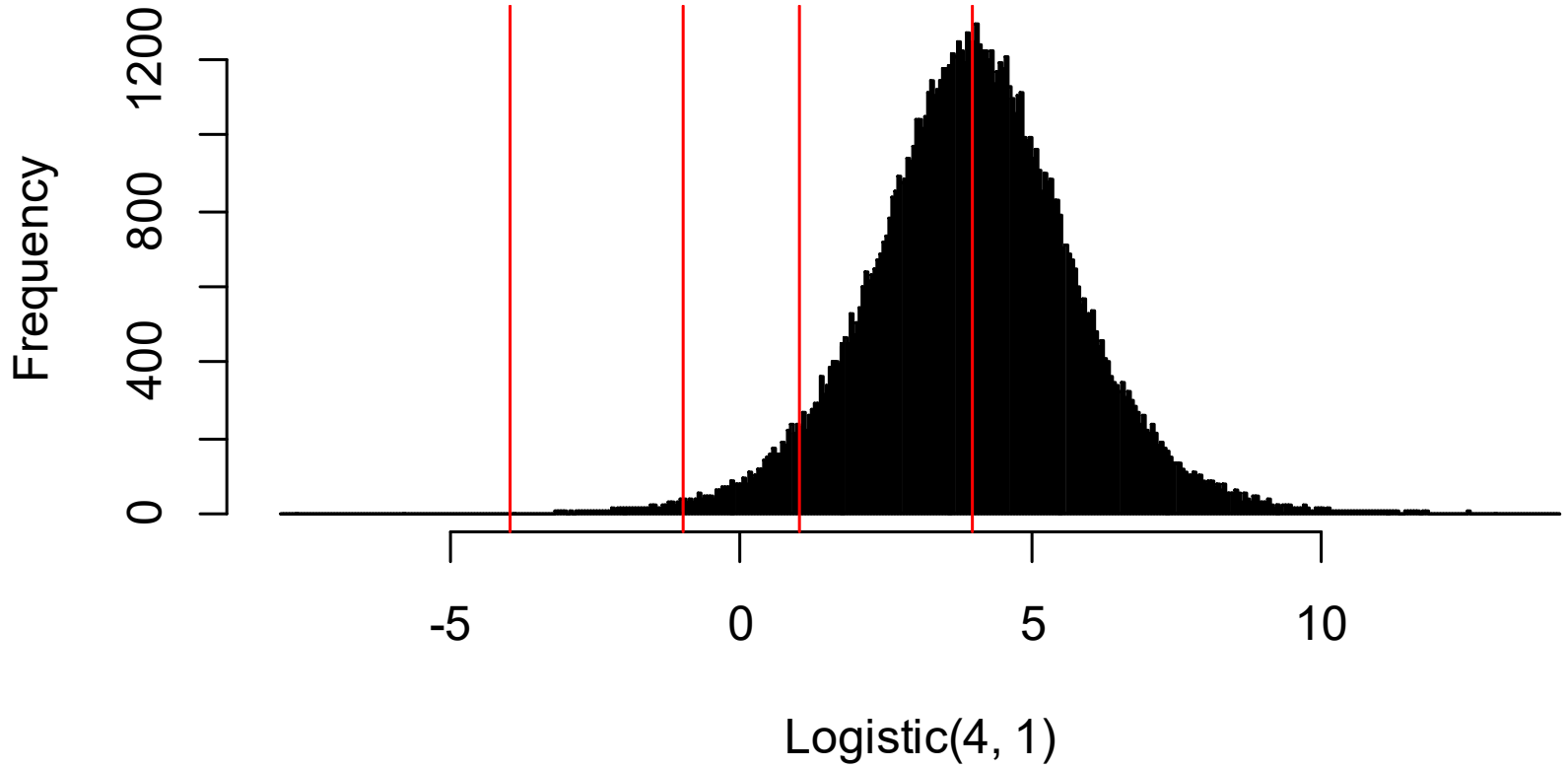


We used ordinal regression to model these data… how's that work…?



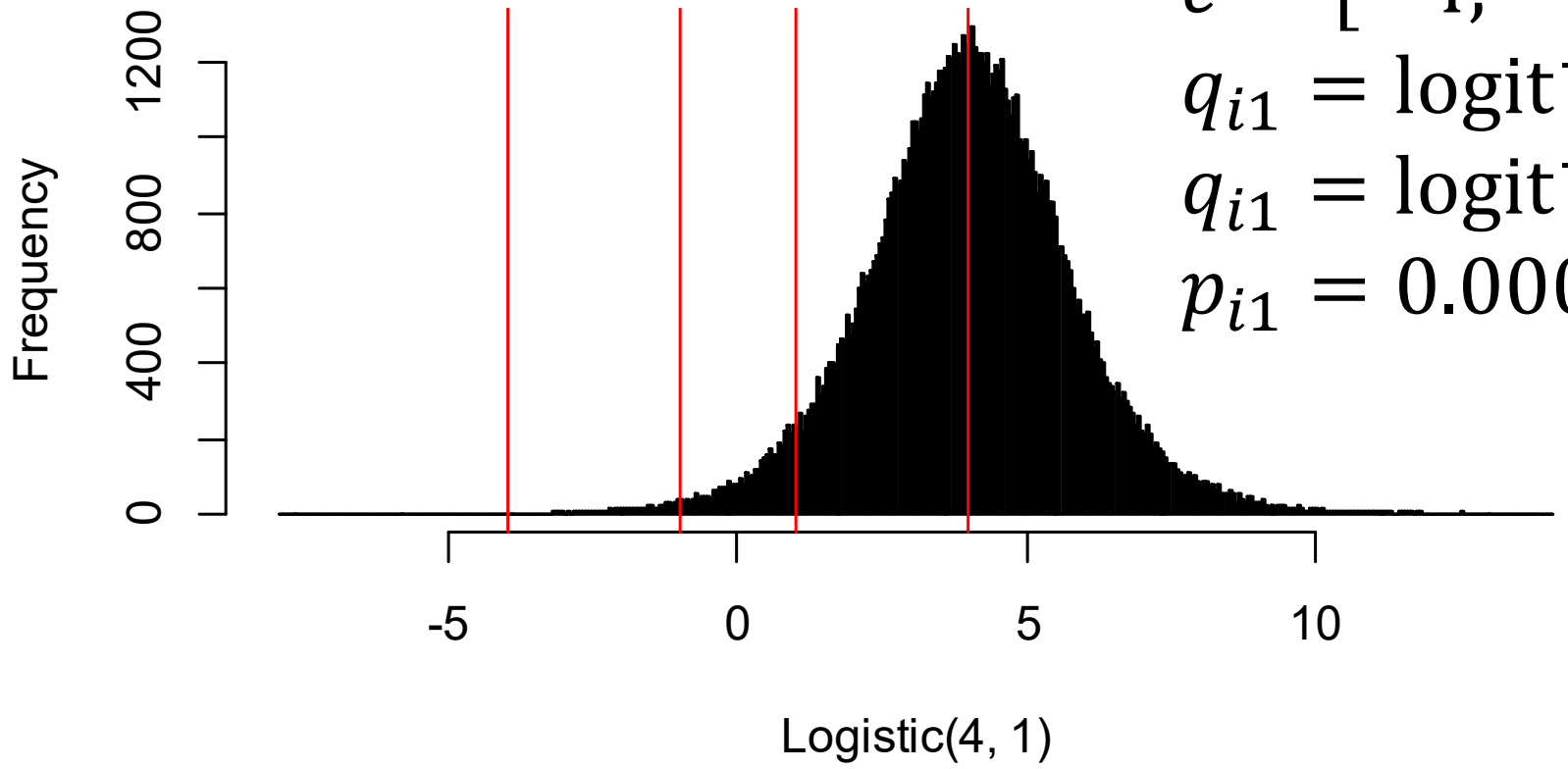
It's really cool! Let's use ordinal logistic regression

Imagine one individual with an attitude of 4 (ε_i=4)



There will be cutpoints at -4, -1, 1, and 4 [we'll estimate these in the model]

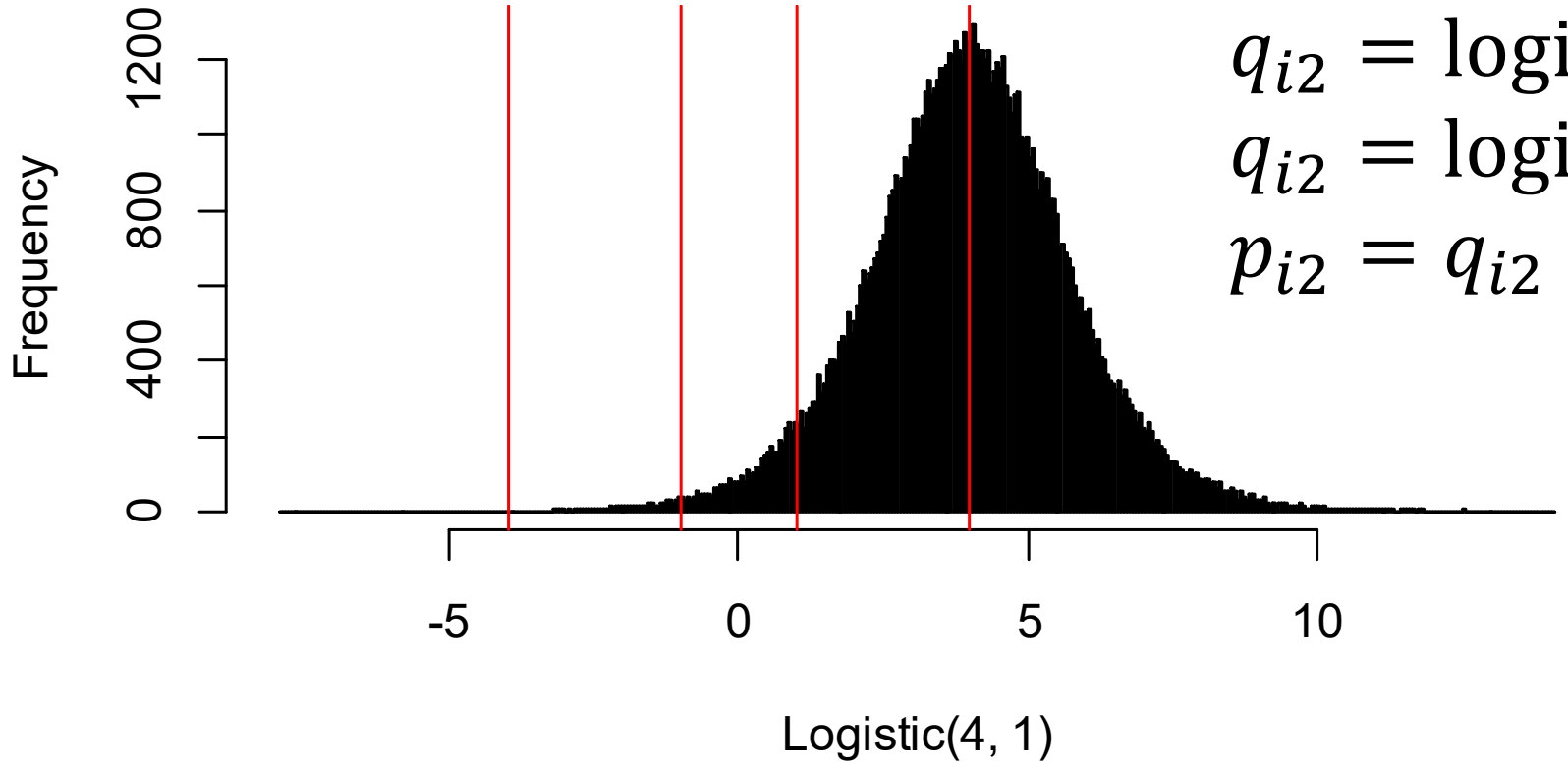
First, we'll obtain the probability of a 1 (strongly disagree)



$$c = [-4, -1, 1, 4]$$
$$q_{i1} = \text{logit}^{-1}(c_1 - \varepsilon_i)$$
$$q_{i1} = \text{logit}^{-1}(-8) = 0.000335$$
$$p_{i1} = 0.000335$$

An individual with a value of 4 is not going to strongly disagree

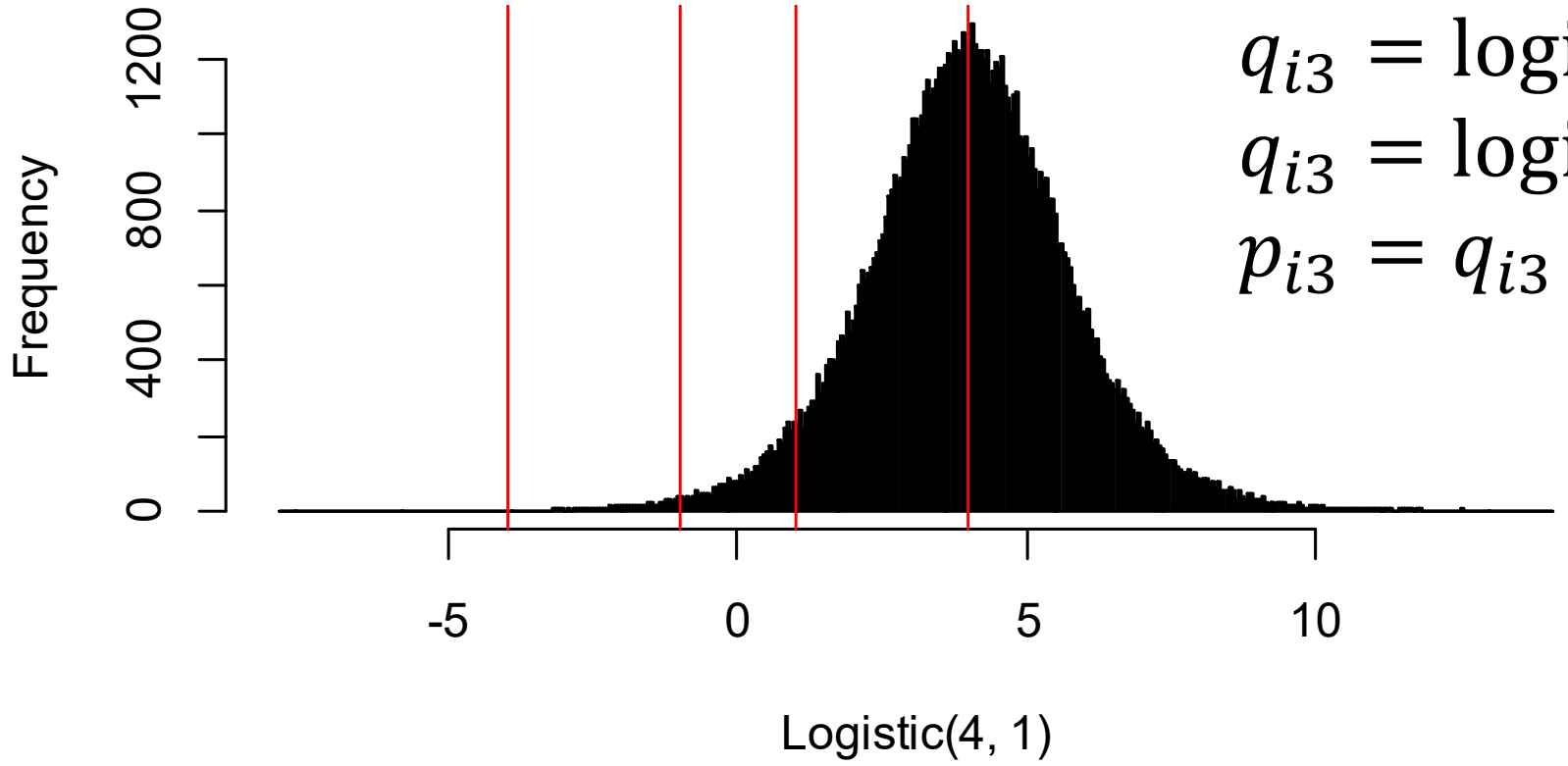
How about a 2 [disagree]?



$$c = [-4, -1, 1, 4]$$
$$q_{i2} = \text{logit}^{-1}(c_2 - \varepsilon_i)$$
$$q_{i2} = \text{logit}^{-1}(-5) = 0.00669$$
$$p_{i2} = q_{i2} - q_{i1} = 0.0063$$

An individual with a value of 4 is not going to disagree

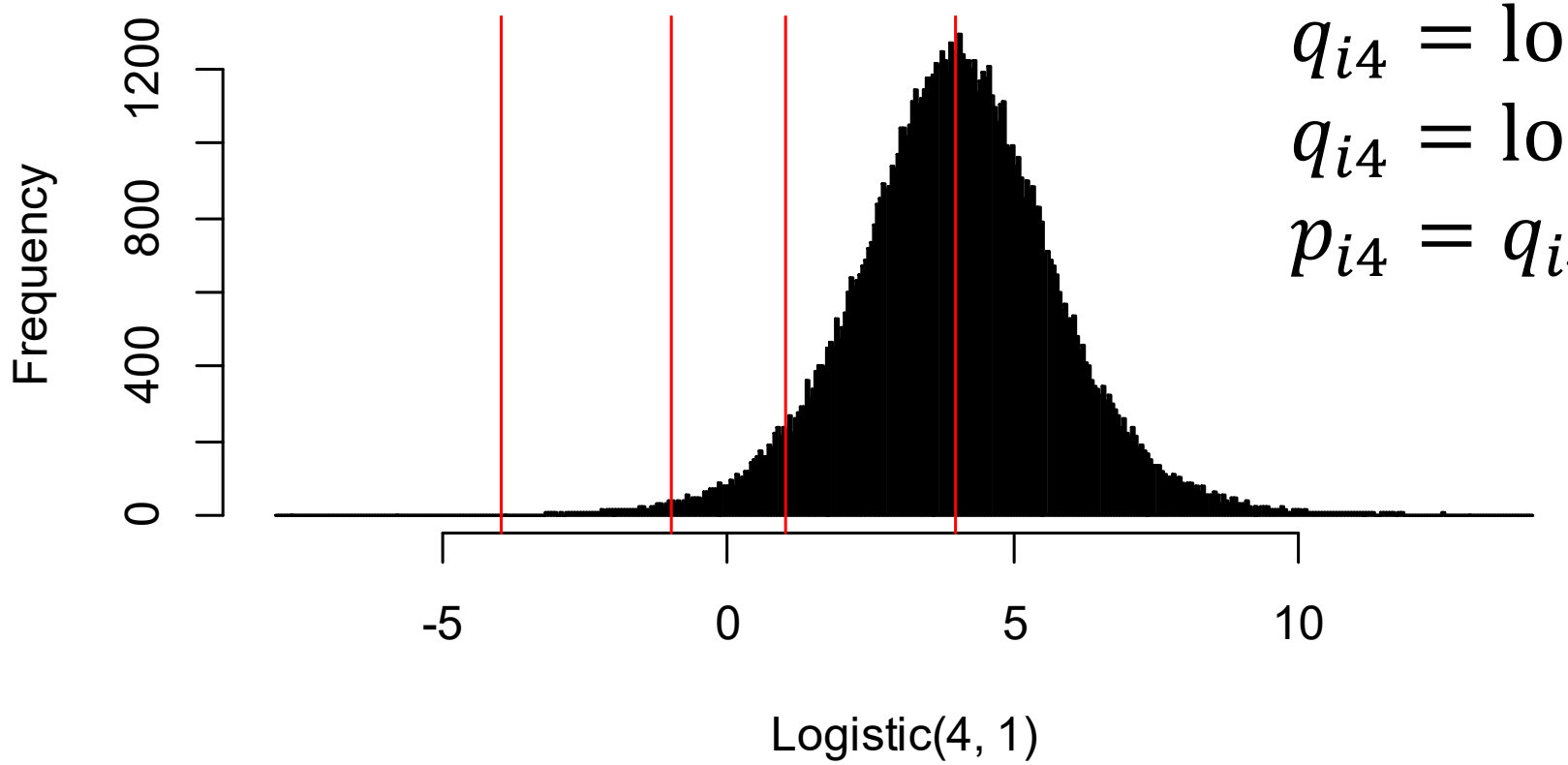
A 3 [neutral]?



$$c = [-4, -1, 1, 4]$$
$$q_{i3} = \text{logit}^{-1}(c_3 - \varepsilon_i)$$
$$q_{i3} = \text{logit}^{-1}(-3) = 0.047$$
$$p_{i3} = q_{i3} - q_{i2} = 0.04$$

They could be neutral (4%)

A 4 [Agree]?

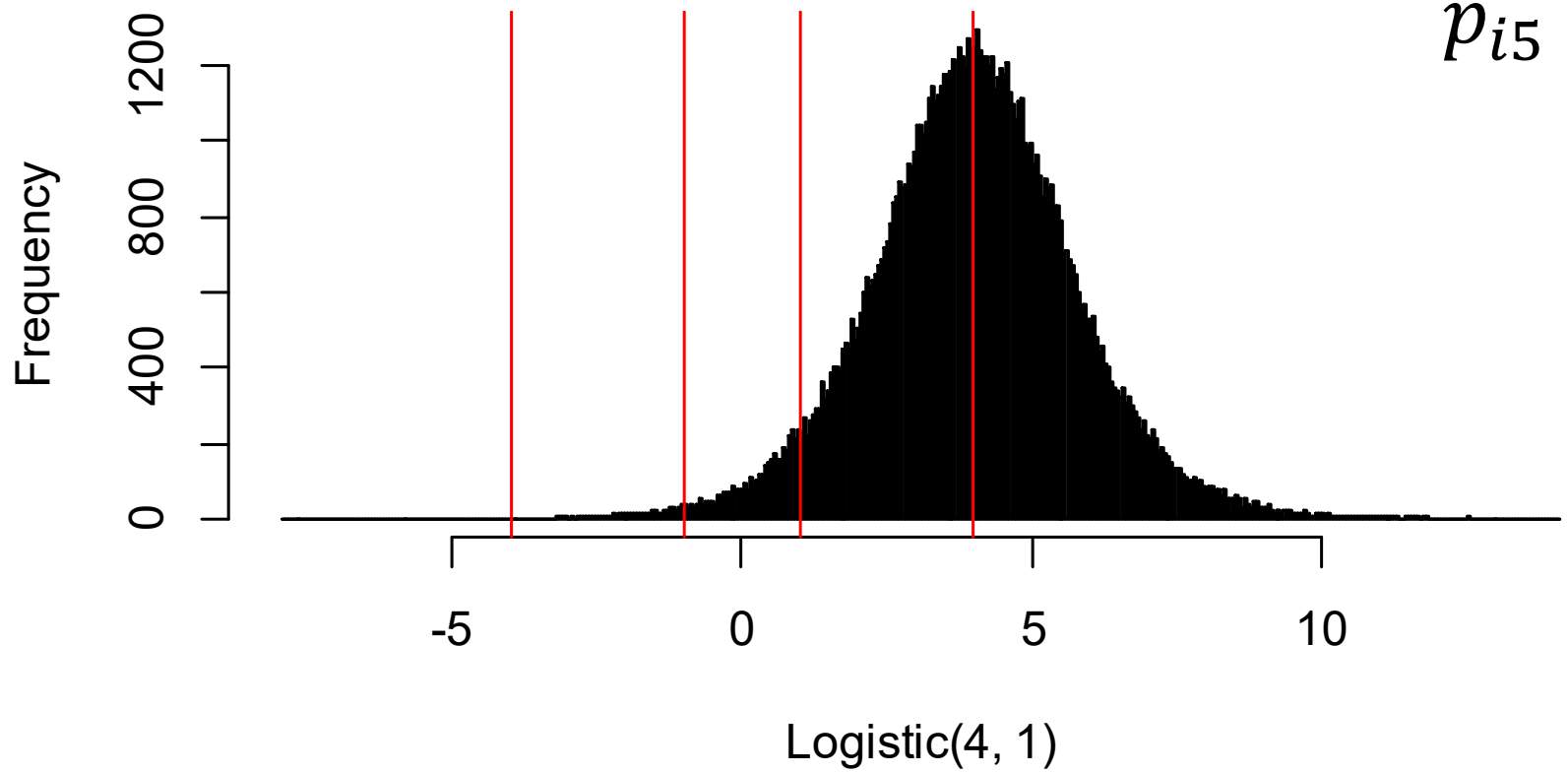


$$c = [-4, -1, 1, 4]$$
$$q_{i4} = \text{logit}^{-1}(c_4 - \varepsilon_i)$$
$$q_{i4} = \text{logit}^{-1}(0) = 0.5$$
$$p_{i4} = q_{i4} - q_{i3} = 0.45$$

They could certainly agree (45%)

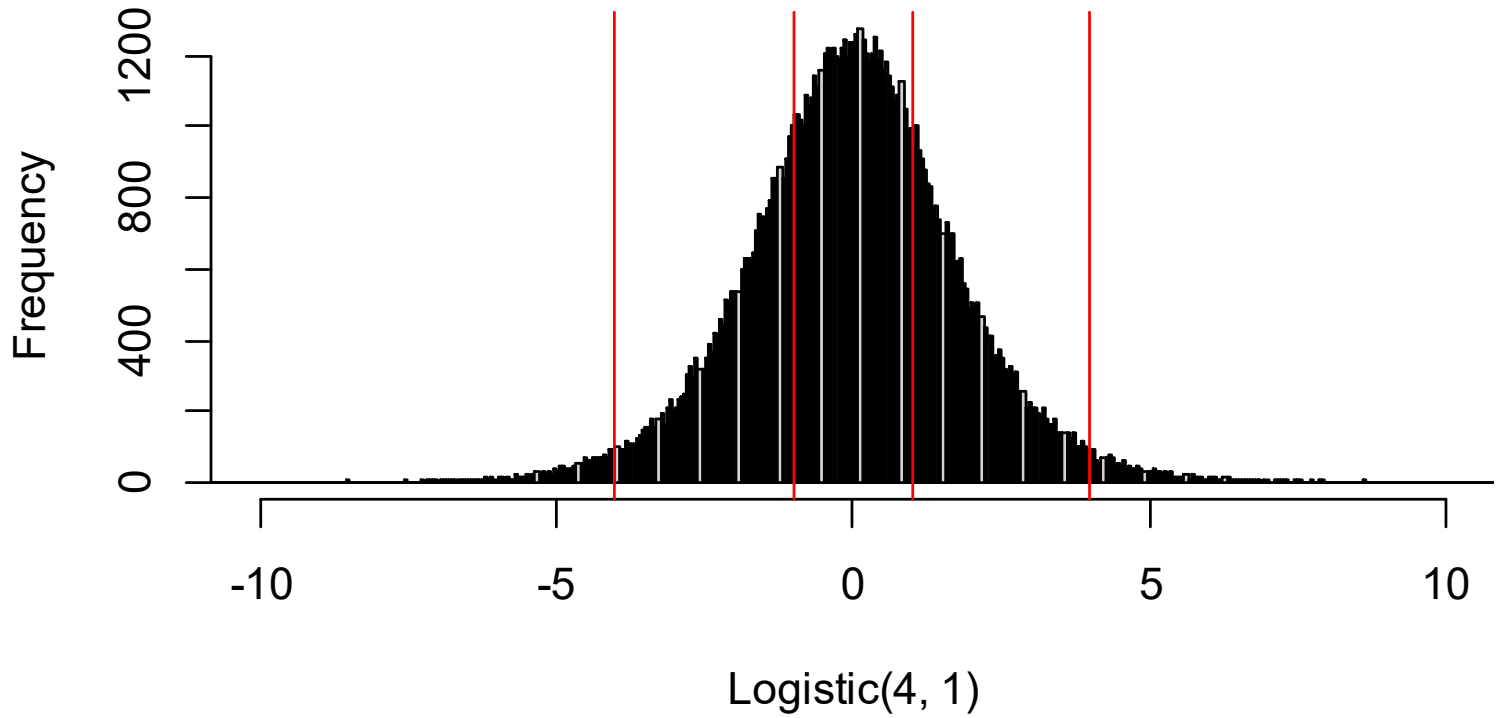
A 5 [Strongly agree]?

$$c = [-4, -1, 1, 4]$$
$$p_{i5} = 1 - q_{i5} = 0.5$$



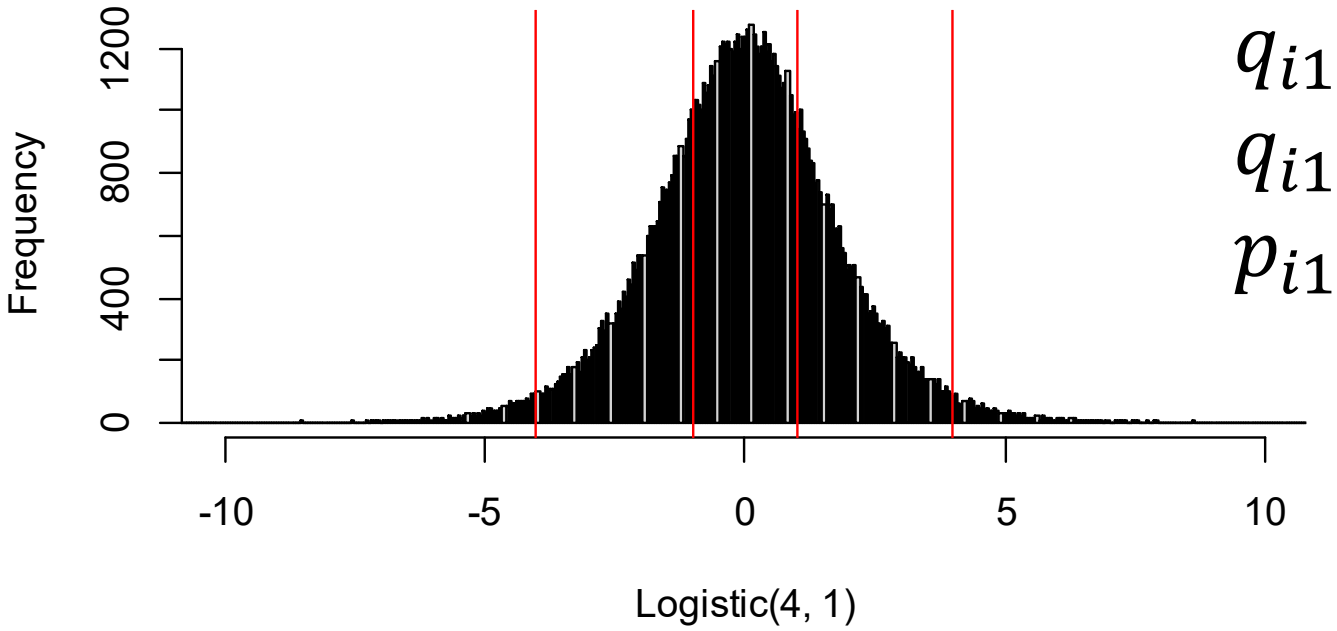
They could certainly strongly agree (50%)

Imagine one individual with an attitude of 0 (ε_i=0)



There will be cutpoints at -4, -1, 1, and 4 [we'll estimate these in the model]

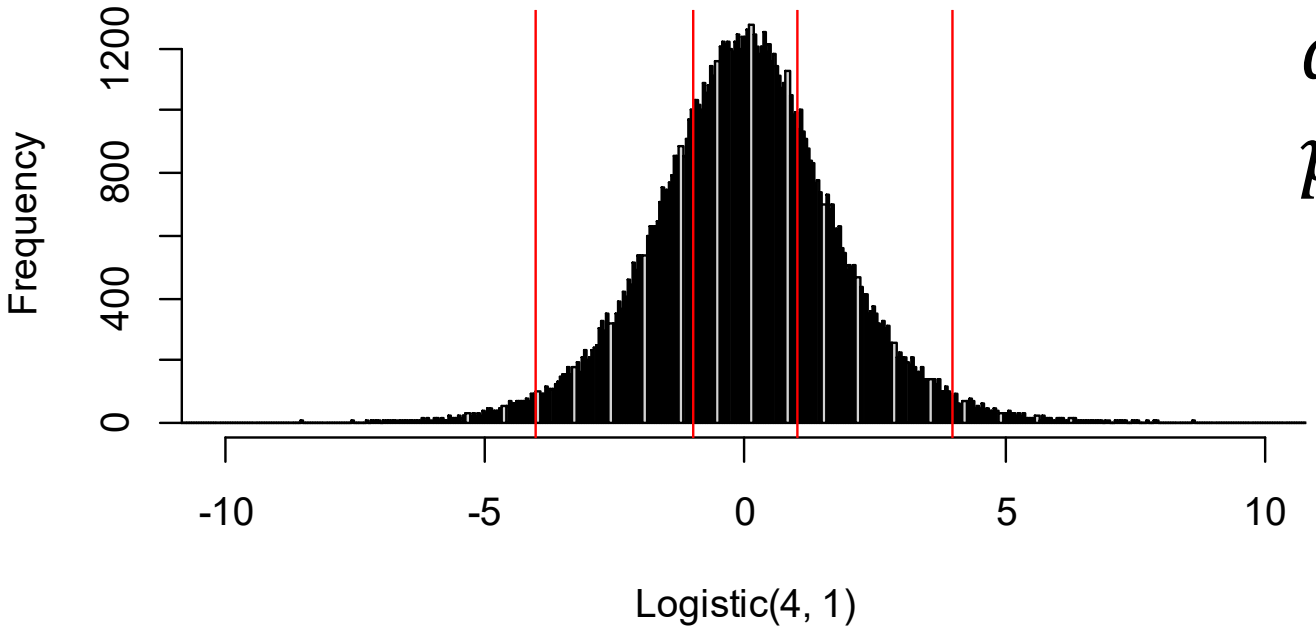
First, we'll obtain the probability of a 1 (strongly disagree)



$$c = [-4, -1, 1, 4]$$
$$q_{i1} = \text{logit}^{-1}(c_1 - \varepsilon_i)$$
$$q_{i1} = \text{logit}^{-1}(-4) = 0.017$$
$$p_{i1} = 0.017$$

An individual with a value of 0 will probably not strongly disagree

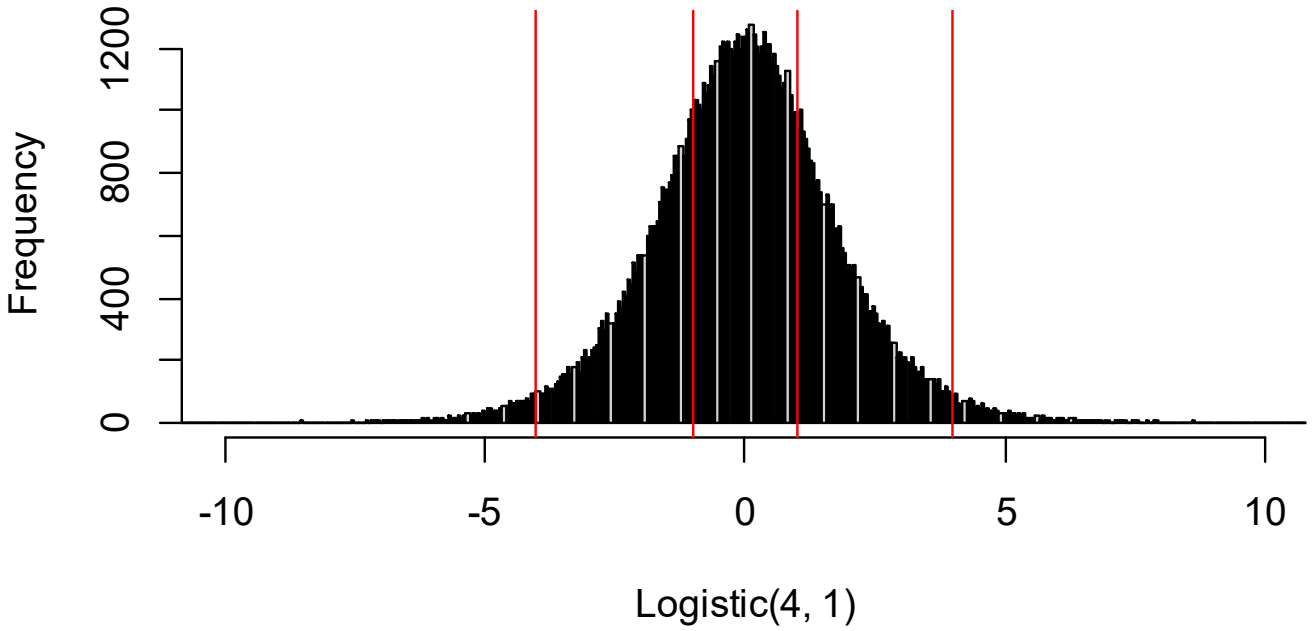
How about a 2 [disagree]?



$$c = [-4, -1, 1, 4]$$
$$q_{i2} = \text{logit}^{-1}(c_2 - \varepsilon_i)$$
$$q_{i2} = \text{logit}^{-1}(-1) = 0.0269$$
$$p_{i2} = q_{i2} - q_{i1} = 0.251$$

An individual with a value of 4 is not going to disagree

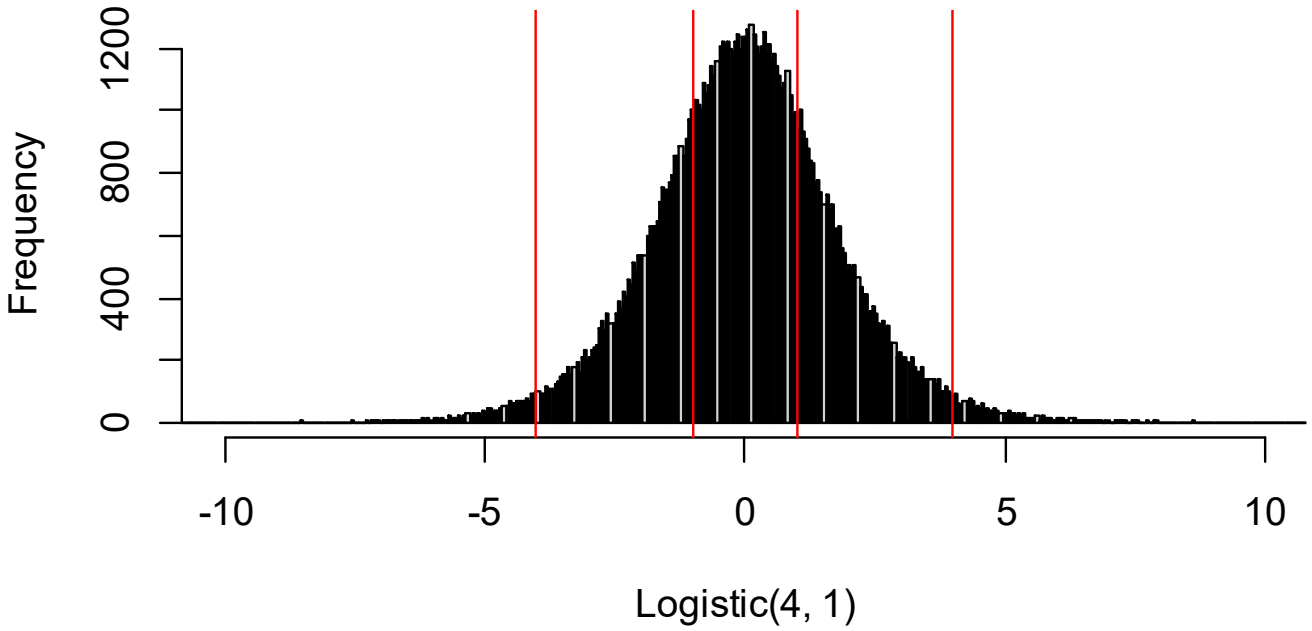
A 3 [neutral]?



$$c = [-4, -1, 1, 4]$$
$$q_{i3} = \text{logit}^{-1}(c_3 - \varepsilon_i)$$
$$q_{i3} = \text{logit}^{-1}(1) = 0.73$$
$$p_{i3} = q_{i3} - q_{i2} = 0.462$$

They could be neutral (4%)

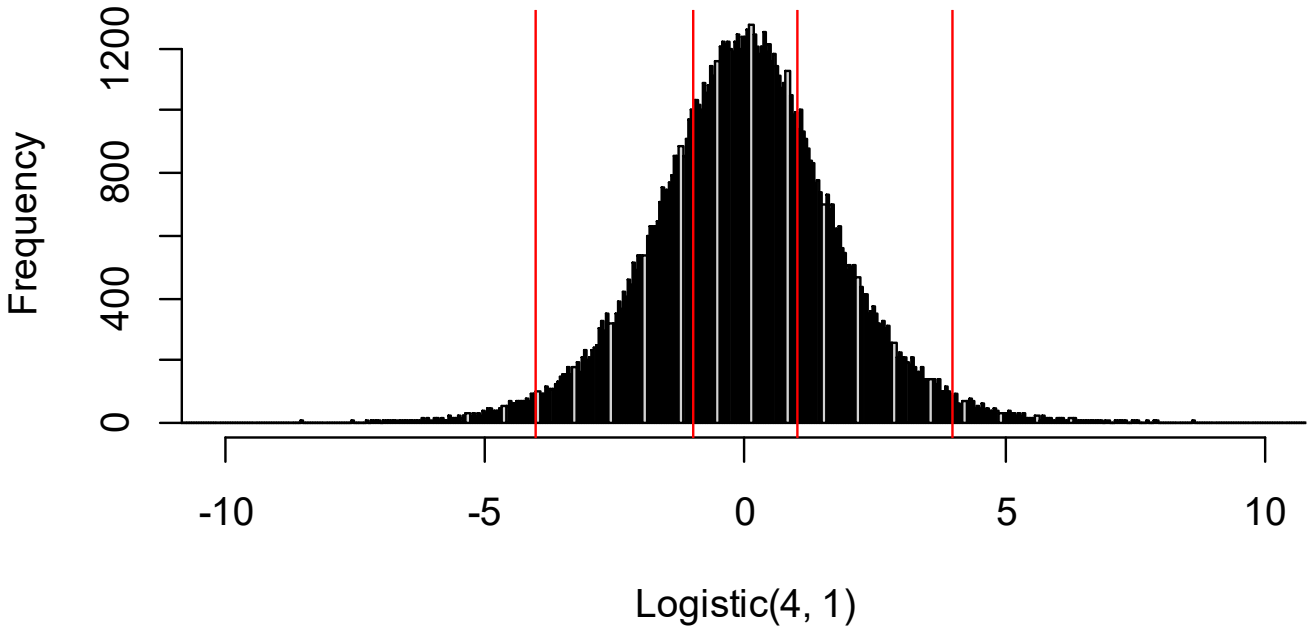
A 4 [Agree]?



$$c = [-4, -1, 1, 4]$$
$$q_{i4} = \text{logit}^{-1}(c_4 - \varepsilon_i)$$
$$q_{i4} = \text{logit}^{-1}(4) = 0.982$$
$$p_{i4} = q_{i4} - q_{i3} = 0.251$$

They could certainly agree (45%)

A 5 [Strongly agree]?

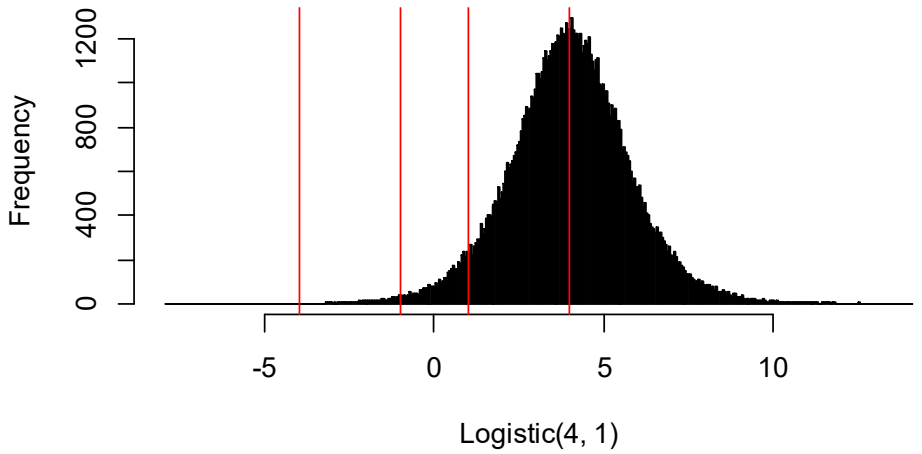


$$c = [-4, -1, 1, 4]$$
$$p_{i5} = 1 - q_{i5} = 0.017$$

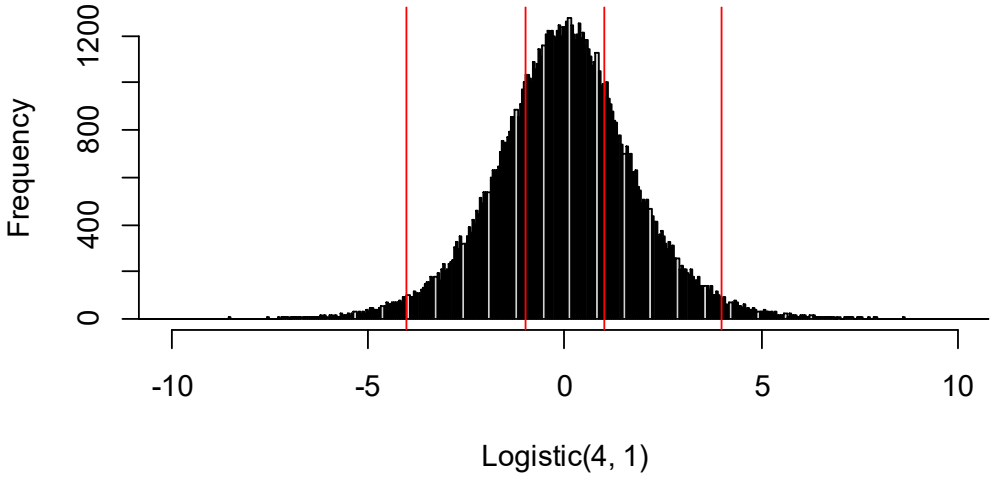
They could certainly strongly agree (50%)

The actual data will be a categorial trial given relevant probabilities

y ~ categorical(0.0003, 0.006, 0.04, 0.45, 0.5)

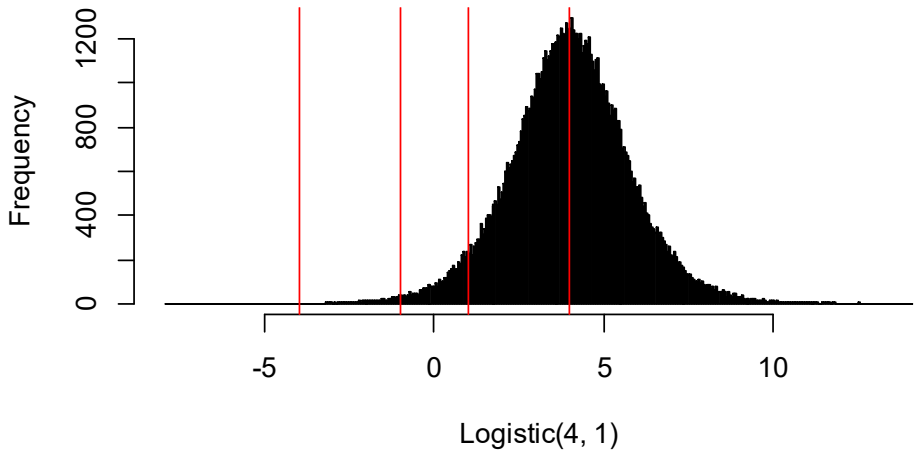


y ~ categorical(0.017, 0.251, 0.462, 0.251, 0.0017)

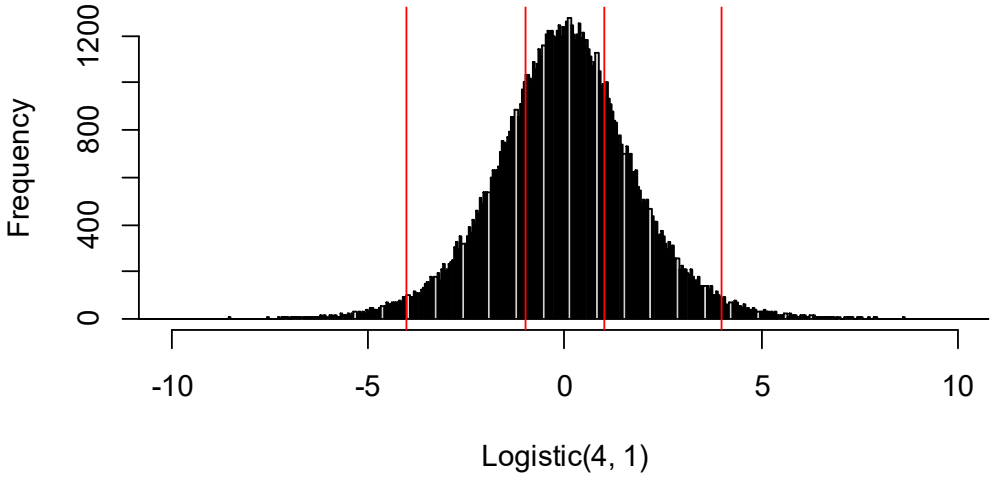


Note probabilities are ‘linked’ and must sum to 1

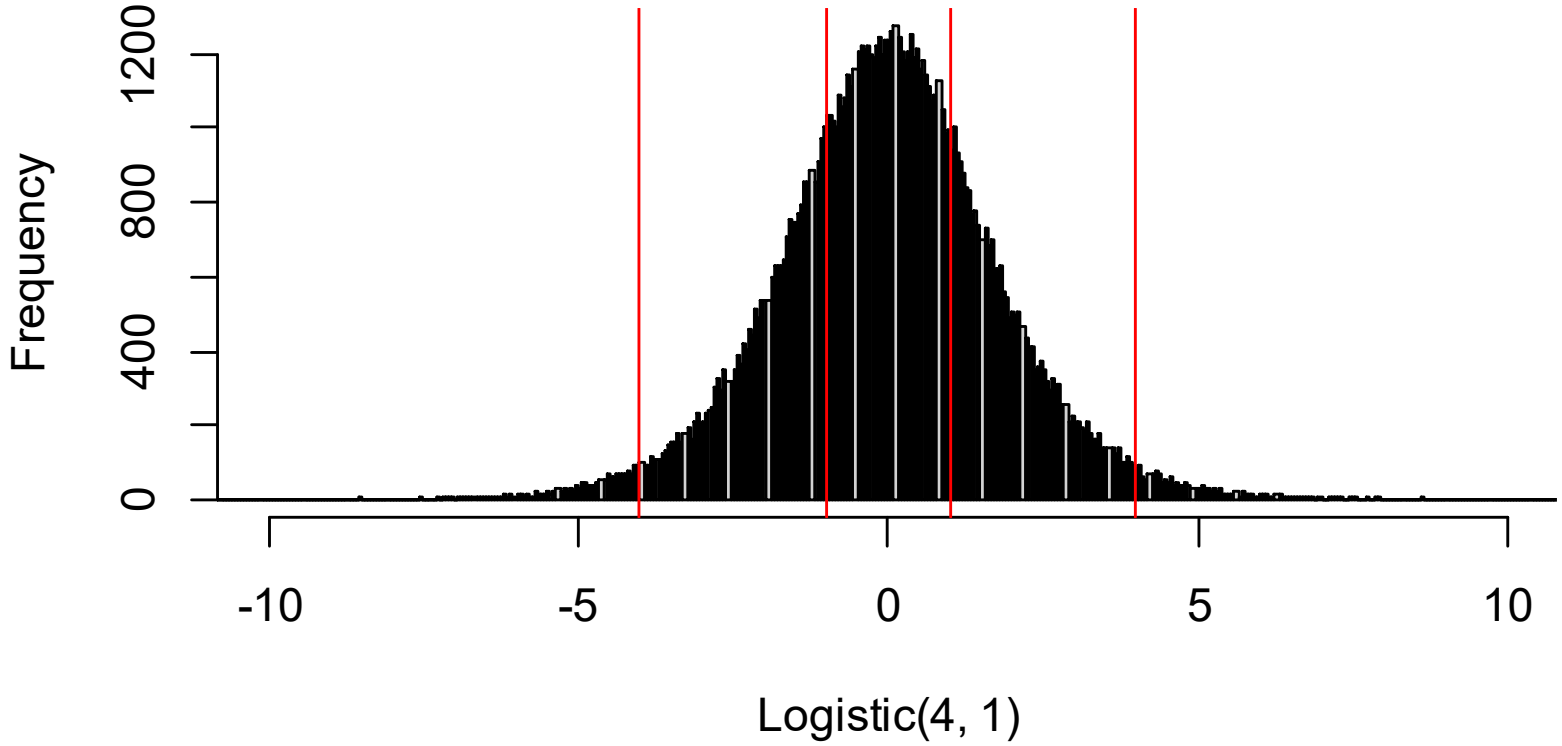
y ~ categorical(0.0003, 0.006, 0.04, 0.45, 0.5)



y ~ categorical(0.017, 0.251, 0.462, 0.251, 0.017)



If strong feelings were rare, and many folks were neutral...



Our cutpoints might look like this...

Bonus simulation (what if attitude affects habitat?!)

Rather than simulating habitat as a Gaussian random field, we'll simulate it as a function of a Gaussian random field and landowner attitude